Spec: Roland SBF-325 Stereo Flanger emulation

Open-source (MIT), non-commercial plugin. Prepared 6 Nov 2026 for an implementing agent. Fidelity is defined by the measurements in Section 8.

1. Purpose and scope

Build an open-source (MIT), non-commercial audio plugin that reproduces the behaviour of the Roland SBF-325 Stereo Flanger (1979–1986): a two-channel analog bucket-brigade (BBD) flanger/chorus with three flanger modes, a chorus mode, feedback, per-channel phase inversion, channel-B modulation inversion, and an external CV input. The deliverable is a VST3/CLAP/AU plugin (plus a standalone test harness) whose measured behaviour matches a reference unit within the tolerances in Section 8. Fidelity is judged by measurement, not description.

Out of scope: emulating the power supply, the footswitch, the input/output level switches as separate gain stages (they are modelled as a single input trim and output trim), and any cosmetic recreation of Roland's panel artwork or trademarks.

2. Reference unit: what is documented

Primary sources are Roland's own Service Notes (first edition, scanned on the Internet Archive) and the Owner's Manual (scan at oldgearguy.com). Roland's current hardware (SPD-SX Pro, SH-4D, V-Stage, MV-8000) contains an in-house recreation of the SBF-325 as an effect block; its parameter set is useful as a cross-check and as a second reference for A/B testing.

2.1 Published specifications (Service Notes)

Item	Value
Delay time	FLANGER 0.5 ms – 15 ms; CHORUS 5 ms – 15 ms
Delay-line frequency response	75 Hz – 14 kHz (±3 dB)
Noise level	< -70 dBm (0 dBm = 0.775 V RMS)
Modulation speed	0.2 s – 40 s per cycle (≈ 5 Hz – 0.025 Hz). Roland's modern block lists Rate 0.02–5.00 Hz.
Input level	H: -16 dBm nominal (0 dBm max); L: +4 dBm nominal (+20 dBm max). Input impedance ≈13 kΩ (service notes) / >100 kΩ noted; verify.
Output level	H: +4 dBm (+20 dBm max); L: -16 dBm (0 dBm max); output impedance < 2 kΩ
Internal rails	±15 V regulated (B3 +14.7…15.4 V, B4 -14.5…-15.2 V) from ≈ ±23–28 V unregulated
Mechanical	2U 19" rack, 482 × 92 × 247 mm, 4.5 kg, 10 W

2.2 Semiconductors in the parts list (Service Notes)

MN3007-class BBD (parts list reads "MN5007" in the OCR; the Panasonic 1024-stage MN3007 is the only plausible reading and must be confirmed against the scan), NE570 compander, µPC4558 dual op-amps, LF555N (one per LFO/clock function), TC4001BP / TC4015BP / TC4079UBP CMOS logic (clock divider / mode logic), 2SK50A FETs (switching/mute), 2SC1815 / 2SA1015 transistors. Trimmers: VR2/VR102 (BBD bias A/B), VR3/VR103 (delay-line gain A/B), VR6/VR106 (clock VCO centre A/B), VR7/VR107 (feedback A/B), VR8 (input indicator), plus LFO trim at CP-1 = 5.00 V ± 0.05 V.

Implication for the model: with a 1024-stage BBD, delay = N / (2·f_clk), so 15 ms corresponds to f_clk ≈ 34 kHz and 0.5 ms to ≈ 1.0 MHz. The BBD's usable bandwidth and its clock-feedthrough/aliasing therefore vary by a

factor of 30 across the sweep; the 14 kHz spec is the long-delay end. This clock-dependent bandwidth is the single most important thing to reproduce.

2.3 Controls and modes (Owner's Manual)

Control	Behaviour to model
EFFECT MODE: OFF	Dry through (effect off, unity with level switches). Separate from BYPASS (hard relay-style bypass, unity gain, unit may be off).
FLANGER I	Monaural: channel-A delay line only; A's output feeds both outputs. CH-B MOD and CH-B PHASE have no effect.
FLANGER II	Stereo: two independent delay lines, A and B. With CH-B MOD = INV the B delay moves opposite to A ("seesaw").
FLANGER III	Cross-mix stereo: both delay lines plus cross-panning of wet signals between outputs (wet A appears in output B and vice versa, mixed with each channel's own). Exact mix coefficients must be read from the schematic and verified by measurement (Section 8.3).
CHORUS	Delay range 5–15 ms, feedback disabled. RATE: CCW lessens effect, CW adds vibrato (i.e. same modulated delay line, different range and wet/dry balance).
FEEDBACK LEVEL	Regeneration around the delay line in FLANGER modes only. Factory trim: VR7/VR107 set so that maximum feedback sits "just before oscillation" across the entire MANUAL range. Model as a hard-limited loop gain < 1 at all delay settings, with the loop's own band-limiting (BBD bandwidth) included.
MANUAL	Sets the centre delay (clock VCO centre frequency). Front panel pot is 100 kA ×2 + 10 kA centre-tapped (EWK2TAK20583) — taper matters for the control curve.
RATE	Internal LFO rate, ≈0.025–5 Hz.
DEPTH	Depth of delay modulation around MANUAL; at 10, MANUAL has no effect (modulation spans the full range).
LFO RATE indicators	Two LEDs: right = positive peak, left = negative peak. Implies a bipolar LFO waveform; waveform shape (triangle expected, LF555-based) is to be confirmed from the schematic and measured at CP-2.
EXT CV IN	External voltage replaces the internal LFO; MANUAL and DEPTH still apply. Model as a sidechain/CV input summed after the depth scaler.
CH-B MOD (NORM/INV)	Inverts the modulation signal for channel B only.
CH A / CH B PHASE (NORM/INV)	Inverts the polarity of the wet signal before summing with dry. INV gives a comb with notches at DC/low frequencies ("flanging weak on the fundamental, strong on overtones").
INPUT / OUTPUT LEVEL (H/L)	20 dB gain steps. Model as input trim and output trim; note the manual's unity-gain pairing (H in / L out or L in / H out).
INPUT LEVEL indicator	Green: signal present; red: ≥ 0 dB (clipping imminent). Calibrated with VR8 at 1 kHz.

3. Signal-flow model

Per channel (A and B are identical; FLANGER I uses only A):

`in → input trim → [compander encoder: NE570, 2:1 compression] → anti-alias LPF → BBD (1024 stages, variable clock) → reconstruction LPF (clock-tracking or fixed; determine from`

`schematic) → [compander decoder, 1:2 expansion] → ± polarity (PHASE switch) → feedback tap (FLANGER modes) back to BBD input → wet/dry sum → cross-mix matrix (FLANGER III) → output trim → out`

Control path: LFO (bipolar, rate 0.025–5 Hz) × DEPTH + MANUAL → [CH-B: optional inversion] → clock VCO (exponential or 1/x law; measure) → BBD clock. EXT CV replaces the LFO term.

Blocks to model explicitly

- **BBD as a variable-rate sampled delay.** Simulate the bucket-brigade as a sample-and-hold chain running at the instantaneous clock f_clk(t): either (a) a fractional delay with clock-dependent lowpass and clock-dependent noise/feedthrough injected (cheap), or (b) a true variable-rate resampler that models the BBD's sample-rate conversion artefacts (Huovilainen/Raffel-style). Start with (b) for correctness, then profile. The delay-vs-control law must be measured, not assumed.
- **Clock-dependent bandwidth.** Model the BBD's inherent lowpass plus the fixed anti-alias and reconstruction filters. Roland's 75 Hz–14 kHz spec is at the slow-clock end; measure bandwidth at 0.5, 2, 5, 10 and 15 ms and fit.
- **Compander (NE570).** 2:1 / 1:2 companding with the NE570's RMS detector time constants (set by its external capacitors; read values from the schematic). This is responsible for the breathing/noise-modulation character and must be in the loop before any feedback nonlinearity is judged.
- **Feedback.** Loop includes the BBD bandwidth and the compander. Factory calibration guarantees no self-oscillation at max feedback anywhere on the MANUAL range; the model must reproduce that ceiling (no runaway) while allowing the hardware's near-oscillation ringing.
- **Modes and matrix.** FLANGER I/II/III and CHORUS as a routing/mix matrix with the coefficients derived from the schematic and confirmed by measurement.
- **Soft nonlinearity.** BBD input stage and op-amp output stage clipping near +20 dBm. Characterise with the THD sweep in Section 8; implement with 2× oversampling only if the measured harmonic content justifies it.
- **Noise.** BBD noise floor (< -70 dBm) and clock feedthrough at high clock rates, shaped by the reconstruction filter. Optional user control ("Noise" on/off) is acceptable but the default must match hardware.

4. Parameter table (plugin)

Parameter	Range / type	Hardware equivalent
Mode	Off / Flanger I / Flanger II / Flanger III / Chorus	EFFECT MODE rotary
Manual	0.5–15 ms (flanger) / 5–15 ms (chorus), displayed in ms; internal pot-taper mapping from measurement	MANUAL
Rate	0.02–5 Hz, log	RATE
Depth	0–100 %	DEPTH
Feedback	0–100 % of calibrated maximum (never ≥ oscillation)	FEEDBACK LEVEL
CH-B Mod	Normal / Invert	CH-B MOD
CH-A Phase, CH-B Phase	Normal / Invert	PHASE switches
Ext CV	Sidechain input or host modulation, ±1.0 normalised, replaces LFO when enabled	EXT CV IN
Input trim, Output trim	-20 / 0 dB steps (plus fine ±6 dB)	INPUT/OUTPUT LEVEL
Bypass	Hard bypass, unity	BYPASS

Parameter	Range / type	Hardware equivalent
Tempo sync	Optional extension; off by default	(none)

5. Reference instruments and data to capture

- A working SBF-325 (used market, roughly US$1–2k) or, failing that, Roland's own recreation inside an SH-4D (≈US$650 new) as a secondary reference; the model should be checked against both, and discrepancies between Roland's recreation and the hardware are themselves worth recording.
- The Service Notes scan (archive.org, item roland_SBF-325_SERVICE_NOTES) for the schematic: BBD part, clock VCO topology, LFO generator, filter component values, NE570 time constants, mode switching, mix resistors.
- Capture set, per channel and per mode, at 96 kHz / 24-bit: (1) swept sine 20 Hz–20 kHz with LFO depth 0 at Manual = 0.5, 1, 2, 5, 10, 15 ms; (2) same with feedback at 25/50/75/100 %; (3) LFO output at CP-2 (DC-coupled) at Rate 0, 2.5, 5, 7.5, 10 and the ±peak LED timing; (4) impulse/burst responses for delay measurement; (5) 1 kHz level sweep for THD+N; (6) silence for noise floor; (7) a 30 s programme-material clip in each mode for null tests.

6. Implementation requirements

- Language/framework: C++ with JUCE or iPlug2, or Faust compiled to C++ for the DSP core (Faust is preferred for the DSP core because the block diagram above maps directly and the compiled code can be unit-tested headlessly). Formats: VST3, CLAP, AU; standalone CLI test harness that renders WAV→WAV for the measurement scripts.
- Internal sample rate: host rate; BBD simulated at its own clock via variable-rate resampling. 2× oversampling of the nonlinear stages only if measurements require it (Section 8).
- Parameter smoothing on Manual/Depth/Feedback (≥ 20 ms) to avoid zipper noise; the LFO itself is not smoothed.
- Stereo processing is two full channels; FLANGER I must truly bypass channel B's delay line (not just mute it) so CPU matches the mode.
- No reliance on the hardware's panel artwork, name or logo in the UI.

7. Repository deliverables

- **README.md** with the credits and trademark/copyright notices in Section 9, a one-paragraph description, build instructions, and the measured-fidelity summary table (numbers, not adjectives).
- **docs/**: this spec; a block diagram; the measurement protocol; a results notebook with plots.
- **measure/**: Python scripts (numpy/scipy/soundfile) for sweep analysis, null test, THD+N, aliasing, noise floor; a **Makefile**/CI task that re-runs them against the standalone renderer and fails the build if any Section 8 threshold regresses.
- **src/**: DSP core with unit tests for each block (BBD resampler, compander, LFO, matrix).
- **LICENSE**: MIT. No Roland material (scans, schematics, firmware, presets) is vendored into the repo; link to the archive.org scan instead.

8. Verification protocol (acceptance criteria)

- Null test against the reference hardware at matched settings: record dry/wet through the same converter chain, align, subtract. Report residual in dB RMS and the residual spectrum. Target: better than -30 dB residual on a 2 s swept-sine and on a drum loop with LFO depth at zero (static settings), degrading gracefully with modulation.

- THD+N vs input level: 1 kHz sine from -40 dBFS to 0 dBFS in 5 dB steps, measured at the plugin output with the effect at a fixed static setting. Plot against the hardware curve.
- Aliasing: 10 kHz and 15 kHz sines at 0 dBFS, maximum modulation depth and rate; any alias component above -80 dBFS is a defect unless the hardware shows the same behaviour.
- CPU and latency budget: report per-instance CPU at 48 kHz / 64-sample buffers on one reference machine; algorithmic latency must be 0 samples (reported to the host as such); oversampling latency, if any, must be reported exactly.
- Every parameter has a documented unit and range in the spec table; the implementation exposes those units (ms, Hz, dB), not 0–127.
- All measurements are produced by scripts checked into the repo (Python, numpy/scipy) that take a WAV pair and emit the plots and numbers. No listening descriptions are accepted as evidence.

Device-specific additions

- 8.1 Delay law: measured delay vs MANUAL position at 11 points, chorus and flanger ranges, must match hardware within 3 % of reading.
- 8.2 Bandwidth law: -3 dB point of the wet path at six delay settings within ±10 %.
- 8.3 Mode matrix: in FLANGER III, feed a signal to channel A only and measure both outputs; the wet/dry/cross coefficients must match hardware within 0.5 dB.
- 8.4 Feedback ceiling: at Feedback = 100 % and any Manual setting, the model must not self-oscillate (output decays after input stops) and the first comb peak gain must match hardware within 1 dB.
- 8.5 Compander behaviour: noise floor modulation during a 1 kHz tone burst (tone on/off) must follow the hardware's envelope within 20 % of the measured attack/release times.
- 8.6 LFO: waveform shape (triangle vs. other) and rate range within 5 % at min/mid/max; ±peak timing visible in the UI meters matches the LED behaviour.

9. Credits, trademarks and licensing (for README)

- "Roland", "SBF-325" and the product's appearance are trademarks/trade dress of Roland Corporation. This project is an independent, non-commercial study of the circuit's behaviour and is not affiliated with, endorsed by, or sponsored by Roland Corporation. Do not use "Roland" or "SBF-325" as the plugin's product name; use them only descriptively (e.g. "inspired by the Roland SBF-325").
- The Service Notes and Owner's Manual are Roland's copyright. The repository must not redistribute the scans; link to the Internet Archive item and oldgearguy.com, and credit the Internet Archive and the anonymous uploaders who preserved them.
- Credit Roland Corporation as the designer of the original circuit (1979).
- Credit any prior open DSP work you build on (e.g. published BBD-modelling papers by Raffel & Smith (DAFx 2010, "Practical modeling of bucket-brigade device circuits") and Huovilainen; cite them in README and in code comments).
- Code: MIT License. Measurement data captured from your own unit may be published under CC0 or CC-BY; say which.

10. Open questions to resolve from the schematic before coding

- Exact BBD part (MN3007 vs. another Panasonic 1024-stage) and whether one or two BBDs per channel.
- Clock VCO law (linear vs. exponential in control voltage) and the 1/x-style delay mapping.
- LFO waveform generator topology and whether DEPTH scales symmetrically around MANUAL.
- Reconstruction filter: fixed cutoff or clock-tracking.

- NE570 external capacitor values (attack/release) and whether companding is in the feedback loop.
- FLANGER III mix resistor values.

11. Sources

- Roland SBF-325 Service Notes (scan): https://archive.org/details/roland_SBF-325_SERVICE_NOTES
- Roland SBF-325 Owner's Manual (scan): https://www.oldgearguy.com/docs/sbf325_man.pdf
- Roland SH-4D effect reference, SBF-325 block (parameter definitions): https://static.roland.com/manuals/sh-4d_v200/en-US/8693773986246923.html
- Roland MV-8000 manual, "Vintage Flanger 325" (notes on oscillation and 1979–86 production): https://www.manualslib.com/manual/142007/Roland-Mv-8000.html?page=416
- Vintage Digital SBF-325 page (overview/specs): https://vintagedigital.com.au/?p=148561
- SMEM collection entry (modes, production years): https://www.smemmusic.ch/index.php/de/node/1761